INVESTIGATING EPIGENOMIC ALTERATIONS IN ALZEHIMER'S DISEASE

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Background and Study Rationale

Alzheimer's disease (AD) is associated with molecular alterations that extend beyond genetic variation, including changes in the epigenomic landscape. Histone acetylation is particularly relevant because it is associated with chromatin accessibility and transcriptional regulation (Klein et al., 2019).

The Religious Orders Study and Rush Memory and Aging Project (ROSMAP) is a longitudinal cohort investigating the molecular mechanisms underlying cognitive aging and Alzheimer's disease (AD), combining detailed clinical characterization with post-mortem brain assessment (Bennett et al., 2018) and extensive molecular profiling (De Jager et al., 2018).

In this study, clinical information was integrated with H3K9ac ChIP-seq data (Klein et al., 2019) to investigate epigenomic differences associated with cognitive status. From **3,584 participants**, **449 individuals with available H3K9ac data were selected (238 AD and 211 NCI)**, comprising **26,384 H3K9ac genomic domains**. Study, sex, and cognitive diagnosis were evaluated as relevant variables for the differential analysis.

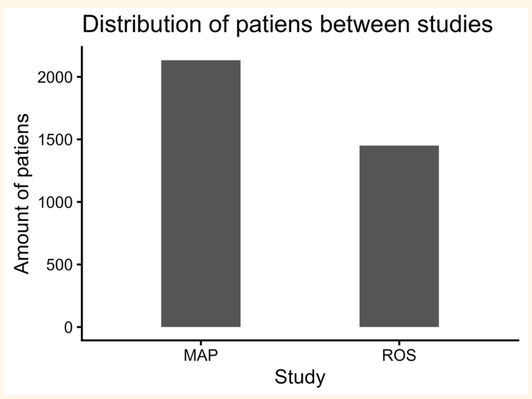
ENCODE RUSH provides complementary experimental data from ROSMAP individuals, enabling broader molecular characterization of the cohort. Here, clinical and H3K9ac epigenomic data were integrated to investigate alterations associated with cognitive status.

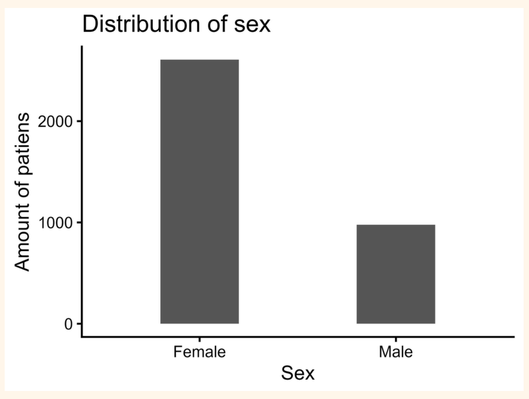
This approach identifies AD-associated H3K9ac alterations while accounting for demographic and cohort-related variation.

Cohort Characteristics

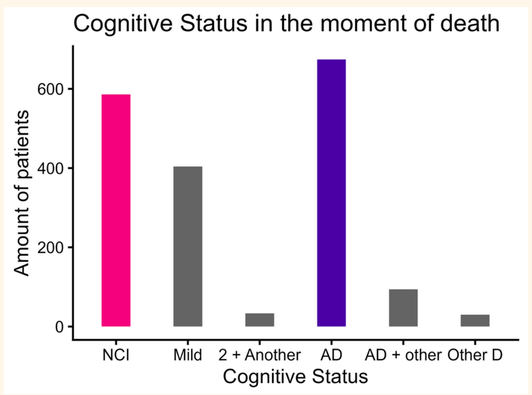
Demographic characteristic		Clinical characteristic	
Group of study	3,583	**Cognitive diagnosis Post Mortem**	1,821
Sex	3,583	**Cognitive diagnosis Pre Mortem**	3,583
Ethnicity	3,583	**Braak stage**	1,580
Education, years	3,582	**CERAD score**	1,597
Spanish ethnicity	3,583	**Age at first AD diagnosis**	832
Post-mortem interval	1,603	**Age at death**	1,855

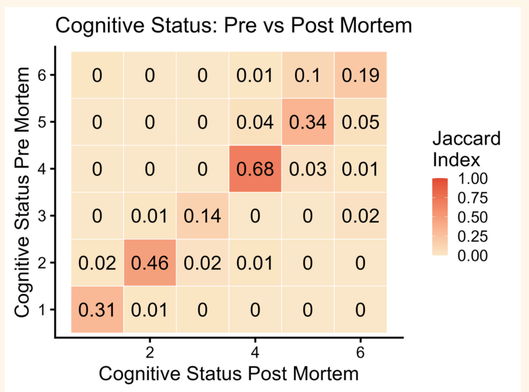
Clinical Variables and Sample Selection





Sex (Male/Female) and Study (ROS/MAP) distributions were assessed as potential sources of variation and included as covariates in the multivariable analysis.



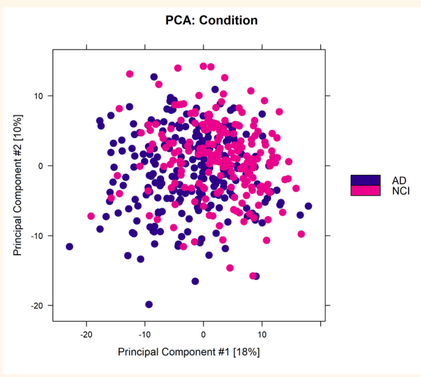


Cognitive status at death was determined through expert neurological review. For differential analysis, No Cognitive Impairment (NCI) was selected as the reference group and Alzheimer's Disease (AD) as the disease group, enabling direct comparison of H3K9ac profiles between both cognitive states.

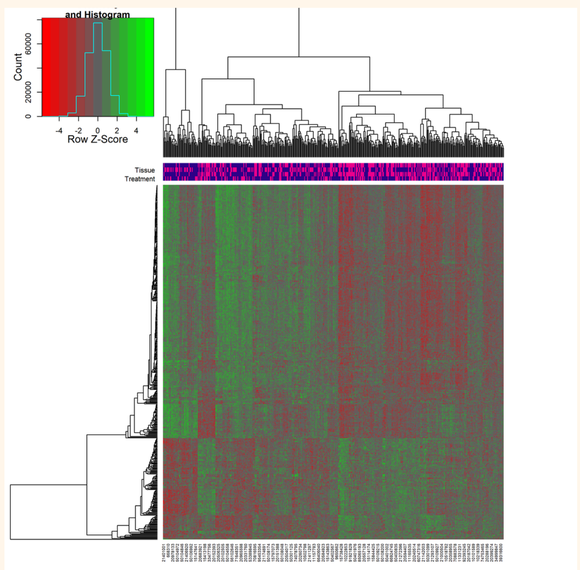
Differential H3K9ac Analysis

Differential H3K9ac domains were identified using DiffBind (Ross-Innes et al., 2012) with the DESeq2 statistical framework. Read counts across 26,384 H3K9ac domains were normalized using Relative Log Expression (RLE) normalization. AD and NCI samples were compared using a multivariable model adjusting for study, cognitive status post mortem and sex. Differential domains were identified using an FDR < 0.05 and subsequently classified as promoter, enhancer, or other genomic regions.

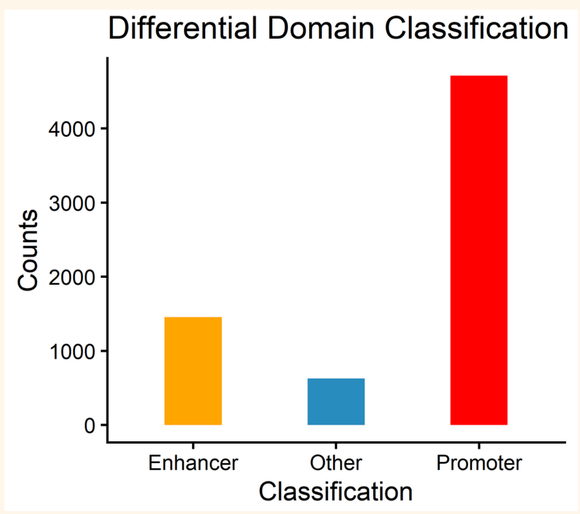
Comparison	AD vs NCI
H3K9ac domains	26,384
Differential domains	6,797
Differential (%)	25.8



PCA was used to assess the global distribution of H3K9ac profiles and the degree of separation between AD and NCI samples for the 6797 differential domains



The top 500 differential H3K9ac domains were visualized across individuals using row-scaled Z-scores, highlighting patterns of relative acetylation between samples.



Differential H3K9ac domains were classified according to their genomic annotation as promoters, enhancers, or other regions.

Together, these analyses characterize the global H3K9ac differences associated with cognitive status and the genomic context of the identified differential domains.

Future Perspectives: Multi-omics Integration

Short term: Validate the promoter, enhancer, and other regulatory-element classifications against the ENCODE 4 cCRE catalog, and determine whether genes associated with differentially enriched promoters are also differentially expressed.

Long term: Incorporate additional ENCODE epigenomic data, including other histone marks, and integrate ROSMAP metabolomic and epigenomic datasets to investigate relationships between epigenetic and metabolic alterations.

References

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